Signal Starvation in GRPO: Measuring and Acting on Zero-Variance Groups in RL Post-Training of LLMs

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Abstract—We study zero-variance groups in short-horizon GRPO-family training. Uniform outcome rewards give zero centred advantages, but a thresholded Zero-Variance Fraction (ZVF) is only a diagnostic proxy. A single-seed, four-arm GSM8K training panel ($G = 8$, 40 steps) gives ZVF 0.71–0.77; three separately stored GSM8K sampling evaluations give pooled 0.1583 and are not the same population. Pairwise contrast density (PCD) is a variance-scale proxy, not a pair-disagreement fraction or a signed mastery signal. The Bernoulli identity $h_G(p) = p^G + (1-p)^G$ assumes conditionally i.i.d. rewards; pooled-rate comparisons and resampling uniform observed vectors do not independently validate it. A live adaptive arm records the same point gain as a fixed recipe at higher rollout cost. A separate offline trace has 92.3% of escalations on observed all-correct groups, not proven true boundary prompts. No controller superiority or general optimum is established.

Index Terms—reinforcement learning, GRPO, large language models, verifiable rewards, advantage estimation, group size

I. Introduction

Reinforcement learning with verifiable rewards (RLVR) trains a language model against a programmatic checker, such as exact match for mathematics or unit tests for code [1], instead of a learned reward model [2]. GRPO [3], [4] has become the default optimiser for this setting because it removes the value network of PPO [5]. It samples G completions per prompt and standardises their rewards within the group:

$$A_i = \frac{r_i - \mu_g}{\sigma_g + \varepsilon},\ \mu_g = \tfrac{1}{G}\textstyle\sum_{j=1}^{G} r_j,\ \sigma_g^2 = \tfrac{1}{G}\textstyle\sum_{j=1}^{G}(r_j - \mu_g)^2. \tag{1}$$

with stabiliser $\varepsilon > 0$. If every completion in the group receives the same reward, $\sigma_g = 0$ and every A_i is zero. Any KL term still acts, but the group-relative part of the update, which is the only part driven by the task, vanishes. Standard training logs (reward, loss, gradient norm, entropy, length) do not show how often this happens: a rising reward curve looks the same whether a fifth or four fifths of the batch is starved.

This paper measures that quantity and asks what, if anything, it tells a practitioner to do. Our contributions are:

- a precise, recomputable definition of the Zero-Variance Fraction and its measurement on real GRPO reward tensors;
- evidence that starvation in our regime is easy-driven, and that ZVF aliases mastery with incapacity, which motivates the pairwise contrast density (PCD) as a more robust signal;
- a signal model for group size with three falsifiable predictions, and the measurements that test them;
- a counterfactual audit of an adaptive group-size controller, together with null results for curriculum and re-baselining interventions.

II. Related Work

DAPO [6] addresses degenerate groups by dynamic sampling, discarding and resampling groups whose rewards are uniform. GSPO [7] moves the importance ratio to the sequence level, and Dr. GRPO [8] removes the length and standard-deviation normalisations that it shows to be biased. Open-source training stacks such as TRL [9], veRL [10] and OpenRLHF [11] implement these variants with different defaults. Other work acts on zero-variance prompts directly, by pre-rollout filtering of prompts predicted to be uninformative [12] or by shaping advantages on zero-variance prompts [13]; we did not evaluate these methods, so our intervention nulls apply only to the heuristics audited here. Henderson et al. [14] showed how sensitive deep RL conclusions are to seeds and implementation details, which shapes our statistical protocol. Unlike this work on new estimators, we do not propose an optimiser; we measure how much of the group-relative signal is available in the first place, and test whether acting on that measurement helps.

III. Method

A. Zero-Variance Fraction

For a batch $\mathcal{B}_t$ of prompt groups at step t, each with G sampled completions and rewards normalised to $[0, 1]$,

$$\mathrm{ZVF}_t = \frac{1}{|\mathcal{B}_t|}\sum_{g\in\mathcal{B}_t} \mathbb{1}\Big[\widehat{\mathrm{Var}}(r_{g,1},\ldots,r_{g,G}) \le \tau\Big], \tag{2}$$

with $\widehat{\mathrm{Var}}$ the unbiased sample variance and $\tau = 10^{-6}$ (distinct from the stabiliser ε in Eq. (1)). Gradient utilisation is its complement, $\mathrm{GU}_t = 1 - \mathrm{ZVF}_t$. For objectives with KL, clipping or auxiliary losses this is an accounting identity, not a claim that the total gradient is zero. Our measured runner has no KL or clip term, so there $\mathrm{ZVF}_t = 1$ guarantees a zero policy-term gradient only when groups are exactly uniform; a positive variance threshold also counts near-uniform groups. For binary

rewards any $\tau < 1/G$ is equivalent to an exact-zero test, since the smallest non-zero unbiased variance is $1/G$. The definition does not apply to dense rewards, $G = 1$, or non-outcome rewards.

B. Pairwise contrast density

For rewards $r_1, \ldots, r_G$ in $[0, 1]$ and $G \geq 2$, define

$$\widehat{\mathrm{PCD}} = \frac{1}{G}\sum_{i=1}^{G}(r_i - \bar{r})^2 = \frac{1}{2G^2}\sum_{i,j=1}^{G}(r_i - r_j)^2. \quad (3)$$

For conditionally i.i.d. Bernoulli rewards its expectation is $(G-1)p(1-p)/G$. This is a variance-scale contrast proxy, not the fraction of distinct pairs that disagree: the latter has expectation $2p(1-p)$. PCD, like ZVF, is unsigned and invariant under $p \mapsto 1-p$; mean reward is needed to distinguish mastery from incapacity.

C. A signal model for group size

For a prompt with per-rollout success probability p and G conditionally independent Bernoulli rewards, the probability that the group is degenerate is $h_G(p) = p^G + (1-p)^G$. We model usable signal per prompt as

$$S(p, G) = p(1-p)\big(1 - h_G(p)\big). \quad (4)$$

The first factor is a heuristic contrast proxy, not a normalised-gradient identity; the second is the probability that any contrast survives centring. Both vanish at $p \in \{0, 1\}$ and peak at $p = \frac{1}{2}$. Three consequences follow: (T1) the expected ZVF indicator of a group equals $h_G(p)$; (T2) for $0 < p < 1$, h_G strictly decreases in G, with asymptotic exponential rate $-\ln \max(p, 1-p)$; and (T3) with unnormalised group-centred advantages, the mean absolute advantage is $2\hat{p}(1-\hat{p})$, so gradient magnitude should track $p(1-p)$.

D. Experimental setup

Training runs use LoRA [15] adapters on Qwen-family models from 0.5B to 32B parameters, together with Llama-3 models, against GSM8K exact-match [16] and synthetic arithmetic verifiers. The training loop logs per-step ZVF, gradient utilisation, entropy, completion length and KL, and persists the raw per-group reward tensors so that every diagnostic can be recomputed rather than read from a log line. The same task, reward and decoding specification is dispatched to a managed training API and to an open trainer; comparisons hold the stack fixed and vary one factor. Panels differ in model, task, seed count and held-out size; we state these per result where the retained artifact records them, and say so where it does not.

IV. Results

A. Starvation is the typical, easy-driven regime

On a stored GSM8K training panel (one seed per arm, 16 prompts $\times$ $G = 8$, 40 steps; GRPO and three variants), recomputing ZVF from the raw per-group reward tensors gives 0.71–0.77 per arm: about three quarters of each rollout batch contributes no group-relative gradient. A separate check recomputes ZVF from three stored Qwen3-8B GSM8K sampling evaluations (200 problems, $G = 8$): 0.130, 0.190 and 0.155 (pooled 0.1583), matching the logged values; these are evaluation samples, not training groups. A matched-budget panel of 2,560 rollouts per arm on Qwen3-8B/GSM8K (two seeds per arm) shows the mechanism: over the last ten steps the $G = 2$ arms reach mean training reward 0.90 and 0.96 with mean ZVF 0.975, while the $G = 16$ arms average ZVF 0.10–0.15 (one step reaches 0.375). When the policy solves the task, nearly every group becomes all-correct. Starvation here is driven by ease, not difficulty.

B. ZVF is a diagnostic, not a score

Two observations argue against optimising ZVF directly. First, it aliases mastery with incapacity. In a 368-run audit (W&B summaries for 350 runs with logged ZVF; nine models; 2–12 steps per run, mostly 5), run-mean ZVF is 0.96 for runs ending at reward below 0.05 ($n = 201$) and 0.90 for runs ending at 0.95 or above ($n = 30$), against 0.17–0.54 in the 0.2–0.65 reward bins ($n = 11$ runs in total). These are very short single-run cells, so this is a descriptive pattern only. Second, ZVF depends on its variance convention. On 600 stored $G = 8$ groups, adding $\delta \sim U(0, 10^{-4})$ to the rewards takes an exact-zero-variance count from 0.158 to 0.000 but leaves the thresholded ZVF ($\tau = 10^{-6}$, unbiased variance) at 0.158, while PCD moves by 3×10^{-7}. Jitter below $\sqrt{\tau}$ cannot cross the threshold (an interval of width 10^{-4} gives sample variance at most 5×10^{-9}); an exact-tie implementation can.

C. Testing the signal model

T1. A group's empirical success fraction mechanically determines whether that same group's binary rewards are uniform. Binning by that fraction therefore is not an independent test of $h_G(p)$. Prompt heterogeneity requires an average of $h_G(p_g)$, not h_G of a pooled empirical rate. The historical theory column below uses unreconciled rate inputs, so it is withdrawn as a quantitative validation; its residual cannot identify dependence or a Jensen gap.

T2. In the 368-run audit, cell-mean ZVF falls across $G = 4/8/16$ for Qwen3-32B ($0.33 \to 0.18 \to 0.08$; $n = 2/2/1$ runs) and Llama-3.1-8B-Instruct ($0.76 \to 0.65 \to 0.47$; $n = 3/1/1$). Llama-3.2-1B, near $p \approx 0$, stays at $0.98 \to 0.97$ ($G = 4/8$), and Qwen3-4B-Instruct is non-monotone (0.89, 0.84, 0.91). The single-run $G = 6$ cells break the ordering for Qwen3-32B (0.42) and Llama-3.1-8B (0.50), so these cells are descriptive, not a test. A controlled Qwen2.5-0.5B sweep on synthetic $a+b$ arithmetic (three seeds per G) gives a clean decline (Table I, Fig. 1), with held-out accuracy flat near 0.98 on 200 freshly sampled problems per run. Raising G buys signal, but on this near-saturated task it does not buy outcome.

TABLE I
Group-size sweep (Qwen2.5-0.5B, synthetic $a+b$ arithmetic, 3 seeds per G, 40 steps, 16 prompts per step): run-mean ZVF averaged over seeds. The earlier theoretical column used held-out accuracy as p and is withdrawn.

G	ZVF (measured)	GU
2	0.8380	0.1620
4	0.7635	0.2365
8	0.6906	0.3094
16	0.6312	0.3688

TABLE II
Four-arm open-trainer audit (two seeds; arm means only). Rollout costs differ. Gains are accuracy-proportion point differences, not equivalence estimates.

Arm	Held-out gain	Mean ZVF	Rollouts
GRPO	+0.500	0.25	120
Dr. GRPO [8]	+0.575	0.27	120
Dynamic sampling [6]	+0.550	0.00	174
Adaptive-G	+0.575	0.23	186

T3. Per-group gradients are not exposed by the managed stack. One logged run of an open backward pass (0.5B model, synthetic arithmetic) reports $\mathrm{corr}(\|\nabla\|, \hat{p}(1-\hat{p})) = +0.71$. Only this scalar survives: no per-group rows, the number of groups or the advantage normalisation are retained. It is directional evidence at most.

D. Group size: no optimum established

The marginal ZVF reduction shrinks with G: 0.0745 from $G = 2$ to 4, but only 0.0594 from $G = 8$ to 16. On the near-ceiling sweep, mean held-out accuracy is 0.982, 0.988, 0.990 and 0.978 at $G = 2, 4, 8, 16$ (3 seeds, 200 problems each). Three seeds do not establish equivalence; the $G = 2$ to $G = 16$ point-estimate ratio is 100.3%. Eq. (4) gives group size a mechanism, but the three-seed sweep establishes no optimum.

E. Acting on the diagnostic

Adaptive group size. We audited an adaptive-G controller that escalates group size on ZVF spikes, using PCD as an aliasing guard, in a four-arm open-trainer comparison (Qwen2.5-0.5B-Instruct, synthetic arithmetic, seeds 0 and 1) with shared data and seeds but unequal rollout costs (Table II). Only arm means survive; per-seed values and the held-out set size are not retained. Dynamic sampling reaches perfect telemetry (ZVF 0.00) at a 45% rollout premium, without beating the cheaper fixed recipe. Adaptive-G ties the best held-out gain at 186 rollouts. This is a tie of reported point gains, not evidence of statistical competitiveness, equivalence or superiority; the comparison is small-n and single-task.

Counterfactual controller trace. The separate 2,560-decision trace includes 1,867 escalation events, of which 1,723 (92.3%) occur on observed all-correct groups and 144 on observed all-wrong groups. The GRPO subset contains 640 prompt-step pairs, 461 uniform (427 correct, 34 wrong). Resampling a finite uniform reward vector necessarily stays uniform: zero restorations at $G = 16$ are a bootstrap identity, not evidence that fresh independent rollouts cannot restore contrast. Observed $\hat{p} = 0$ or 1 does not establish a true boundary probability. The trace is not the live four-arm experiment and supplies no measured controller gain.

Curriculum and re-baselining. A difficulty curriculum that filters collapsed groups drives the zero-gradient waste fraction from 0.50 to 0.00, but at 4.81× the sampling cost, and its held-out gain equals the baseline's (+0.05 in both arms; one 8-step run per arm). At a matched 30k-token budget the two tie at a mean +0.028 over three seeds per arm. A single-seed $G = 4$ "win" of +0.125 (held-out 0.875 to 1.000) did not survive re-testing: in a three-seed group-size panel the $G = 4$ gain is 0.0 in every seed. Re-baselining is not specified sufficiently here to assess its estimator or gradient effect. Equal reported point gains in these panels do not establish that interventions cannot improve held-out outcomes.

V. Discussion and Limitations

These are short, heterogeneous telemetry panels, not a single confirmatory experiment. The 0.71–0.77 training-panel values and the stored sampling-evaluation checks (0.1583 pooled) have different cohorts; their difference is not a replication discrepancy resolved by matching a log. The exact main-panel configuration, intervention definitions, seed/item counts, control law and uncertainty records are incomplete here. Equal reported point gains do not establish equivalence. The live four-arm audit uses unequal rollout costs (120, 120, 174 and 186), and is separate from the offline controller trace. No matched-cost superiority follows. The heuristic signal model is not an expected normalised-gradient identity. T2's strict decrease holds only for $0 < p < 1$, with the stated rate an asymptotic exponential rate; at true boundaries $h_G = 1$. Toy gradient correlation lacks a fully specified normalisation, norm and sampling unit and is descriptive only. Dense rewards and positive variance thresholds require their own calibration. No fixed operational alarm threshold is validated by this evidence.

VI. Conclusion

ZVF and variance-scale PCD can describe reward contrast, but both are unsigned and require mean reward and source-bound configuration alongside them. The historical sweep shows declining measured ZVF with G; it establishes neither an outcome optimum nor independent validation of the pooled-rate model. Uniform-vector bootstrap failures are mechanical, not fresh-rollout impossibility results. A pre-registered, fully specified, matched-cost controller comparison remains future work.

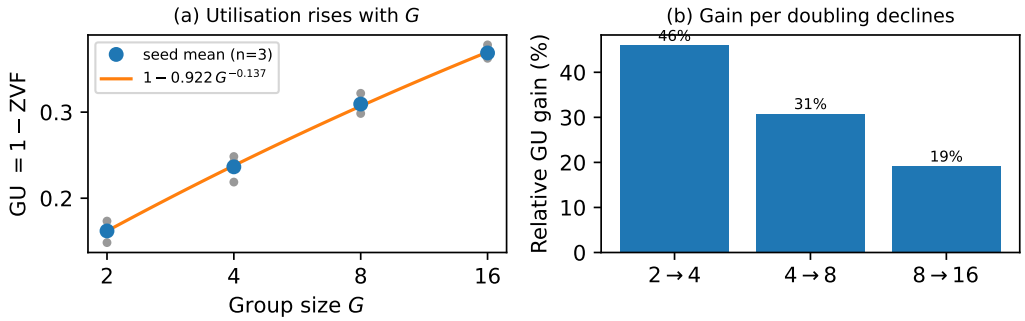


Fig. 1. Gradient utilisation ($1 - \mathrm{ZVF}$) in the Table I sweep rises with group size with diminishing returns (left; grey = seeds; curve = power-law fit $\mathrm{ZVF} = 0.922\,G^{-0.137}$, $R^2 = 0.9996$ in log space on the four means); the relative gain per doubling of G declines (right: 46%, 31%, 19%).

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